

Product Requirements Document

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Revision A

**For Modularm Controllers
Plus (+) & Pro**

Approvals

Kitchen Brains:

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Revision History

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1 GENERAL DESCRIPTION

The purpose of this project is to develop a walk in refrigeration controller product line. This will be done in 2 Phases.

We will leverage all the current features and functions of the 75LCT but extend this to a controller by controlling the various components in the walk in. For the scope of the project this document highlights the additional requirements to support this. Refer to the existing 75LCT PRD for reference in existing operation.

Base 75LCT functions are:

- Touchscreen Platform
- Patented Light Delay Control/Extended Open-Door Alarm
- Critical Alerts
- Elevated Temperatures
- Clean Condenser Alert (Run Time)
- Check Evaporator Alert (Run Time)
- Panic Alarm, IP-1 functionality
- Motion Sensing, MD-1 functionality
- Universal Power Supply 100v-240v 50/60Hz – Global Compliance
- Zero Crossing Circuit (Relay output)
- Optional Lead/Lag Configuration (2 Systems)
- Cloud Monitoring/Alerting- Modularm Dashboard

The following additional features and I/O will be in the new Modularm Plus Controller (as an addendum) – Phase 1

- Real Time Clock (RTC)
- Compressor Control
- Evaporator Fan Control
- Defrost Control
- Door Heater Control
- Humidity Sensing & Reporting
- Rapid Cool Logic (when connected to Lead-Lag Systems)

In addition to all the above, the following additional features will then be included in the Modularm PRO Controller – Phase 2

- CO2 Sensing
- Electric Expansion Valve Control (EEV)
- Humidifier Control
- Refrigerant Leak Detection
- Vent Output
- Future AUX output
- Future Serial Communications for “add-on” boards/functions.

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By adding these features in a “Plus” model, KB can then market and offer:

Features:

- Centralized control (replaces thermostat, defrost timer, fan delay timer)
- Adaptive control logic
- Manages the defrost cycle, eliminates ice/frost formation
- Rapid cool options (enables both refrigeration systems)
- Programmable time/temp interval
- Visual and audible alerts
- Cloud monitoring, alert options, and reporting
- OEM and retrofit options

Benefits:

- Simplifies wiring & complexity
- Reduces defrost heat requirements
- Prevents freezer burn due to excessive temp swings/defrost cycles
- Monitors Employee Safety
- Maintains highest food / product quality
- Saves energy

And with these additional features for the “Pro”:

- CO2 sensing, monitoring, and excessive level alerts
- Humidifier output monitor and control humidity level
- Refrigerant leak detection and motorized vent output control

Benefits:

- Automatically opens vents when leaks are detected, or excessive CO2 levels are reached
- Full refrigeration control, monitoring, and alerting

2 REFERENCES

- 75LCT Manual & Spec Sheet
- 75LC Manual & Spec Sheet
- DAC55 – For Evaporator Fan Interrupt
- 75LCT PRD UPDATE EJM Sept 15 2020
- Modularm Admin App (for Bluetooth Firmware reflashing)
- VOC Modularm Next Gen PPT

Similar Competitive Offerings

- Thermokool TK4700 Controller Spec sheet (and ref Youtube Videos)
- Everidge Smartrite SMRT-100 Controller Manual (and ref Online docs) <https://everidge.com/smartrite-helpcenter/index.html>
- IntelliGen Control - [intelliGen Refrigeration Controller by Heatcraft- Simplicity Refined](#)
- KE2THERM - [KE2 Combo Display - Acts as temp display, lighting control, panic alarm, temp alarm & door controller. Use stand-alone or with 1 or 2 KE2 Evap OEM controls.](#)
- <https://www.ke2therm.com/products/detail/947918/ke2-therm/ke2-evap-oem/evaporator-control-that-saves-energy-maintains-tight-temp-control-has-built-in-communications-leadlag-control-humidity-options-more>
- Kolpak Artic Fox (old & current models)

Refrigeration Controller - Logic References

- Carel

- Danfoss
- Bally Lead/Lag Control Panel Models BLC1/BLC2/BLC3

3 HARDWARE REQUIREMENTS

Requirement Number	Description	
PWRIN-00	3.1 POWER INPUT	
PWRIN-10	The control will support global 100-277VAC power input requirements +/- 10% line, 50/60Hz, fused input as required with commonly available fuses available to the technician.	
PWRIN-20	The control will not exceed XXXX when under full I/O load with optional communications.	
PWRIN-30	The control will support a safe shut down during a brownout or power interruptions.	
PWRIN-40	There will be a battery backup option to support the 5in display for a minimum of 30 minutes to annunciate and alert of a power failure condition while briefly illuminating the display periodically of the event and keeping the IP-1 or IP-2 illuminated to allow a user to safely exit the walk in.	
PWRIN-50	Unit needs to support current SLEEP function for shipping from both the KB factory, as well as from the OEM to the installation side, to avoid battery discharge during transportation and inventory stocking/storage. This should be accessible from the front panel and can be chosen an unlimited # of times for the reasons mentioned above.	
INPUT-00	3.2 INPUTS – ANALOG & DIGITAL	
INPUT-10	There will be (4) 10K Thermistor Inputs for: Walk In air temp, Coil 1, and Coil 2 Temps, and Heater 2 Input. With accuracy of 2 degrees F.	
INPUT-20	There will be 2 Door Switch Inputs. Door switches can be wired in serial or parallel. In system programming the user can define 1 or 2 switches for logic.	
INPUT-30	There will be a humidity sensor input (0-5v analog?) – KB eng to confirm most common off the shelf signal/devices. This can also be an integrated humidity & temp sensor to take the place of 1 thermistor probe from INPUT-10	
INPUT-40	The unit will support the current IP-1, IP-2 Assemblies for Panic and Light Delay Function (ref KB 231-40012-XX & 231-40015-XX).	
INPUT-50	The unit will support up to (3) MD-1 Assemblies for Motion Detection Logic (ref KB 1 231-40013-XX). Locations can be outside or inside the walk in itself, operation will remain the same as currently.	
INPUT-60	The unit will support a common interface to 3 rd party CO2 sensors as inputs for CO2 level detection.	
INPUT-70	The unit will support a common interface to 3 rd party Refrigerant Leak sensors/devices as inputs for leak detection.	
INPUT-80	The unit will support EEV valve positioning. Ref section 8.2	
INPUT-90	The unit will support a common interface for a pressure transducer. Input 0 to 5 Vdc - Pressure Range: 0 to 150 psia - Proof Pressure: 450 psi - Burst Pressure: 1500 psi Operating Temperature: -40°F to 275°F (-40°C to 135°C)	
INPUT-100	Inputs will have any necessary circuit protection as required.	

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OUT-00	3.3 OUTPUTS	
OUT-10	There will be a Relay Output for Light Control – upgraded from current 8Amps on 75LCT to 15Amps rating VOC requested, zero crossing.	
OUT-20	There will be a 10Amp Relay Output for remote Compressor Relay/contactor Coil Control	
OUT-30	There will be a (20Amp) Relay Output for the Evaporator Fan for fixed speed. This will be decided in system programming	
OUT-40	There will be a (20Amp) Demand Defrost Output	
OUT-50	There will be (2) 10 Amp Relays used for Lead Lag System operation (2 nd part number) that allows both refrigeration systems to energize electrically at the same time for “Rapid Cool” logic, when enabled. These need to be independently controlled and wired for individual control.	
OUT-60	There will be an analog output HEATER 1 to control a remote SSR to vary current to a door heater (heat tape). The reason is doors can come in various sizes so the length of heat tape can vary based on the unit. Actual Load can be 9-12 watts per foot, typically 288 watts for remote SSR, Modularm control can control via the analog output.	
OUT-70	There will be a set of dry relay contacts (Alarm) that will energize the relay if the control is in an error or alarm condition. This can be then wired to a Central Station alarm panel should the end-user desire.	
OUT-80	There will be a (10Amp) Heater 2 Output that is switched via a relay to control a viewing window or “port style” window and will either be temp based or duty cycle.	
OUT-90	There will be a (10Amp) Relay output to control the power vent output	
OUT-100	There will be a (10Amp) Relay output to control the humidifier output	
OUT-105	The unit will support EEV output positioning. Ref section 8.2	
OUT-110	Relay outputs should have a provision for snubbing loads as/when required.	
OUT-120	Unit will support options for a variable speed fan using a 0-10V output - according to California Title 24 regulation as a reference. This will be decided in System Programming.	
DISPLAY-00	3.4 USER INTERFACE DISPLAYS	
DISPLAY-10	This will utilize a 5in PCAP Color Display in a portrait orientation.	
DISPLAY-20	The UI flow and screens will follow the same look as the current 75LCT but with the larger display.	
DISPLAY-30	The SYSTEM A & SYSTEM B Led indicators for lead lag will be incorporated into the 5in display with icons.	
DISPLAY-40	There will be a branding logo added at the top of the home screen.	
COM-00	3.5 COMMUNICATIONS	
COM-10	Control will support Bluetooth for OTA firmware updates. When pairing to a device via Bluetooth App. the 75LCT Plus & Pro should have a user acknowledge the pairing on the device. There will be a pairing process similar to the current ModularmAdmin app.	
COM-20	Control will have a local USB socket.	
COM-30	Control will support Wi-Fi as a standard offering, and send all content via REST JSON API messaging or like standard format to KorrektTemp dashboard. Standard Wi-Fi security methods for current and future requirements will apply. Updated methods will either be upgraded wirelessly or via USB if required. When WiFi is connected there will be a WiFi icon illuminated on the home screen to show network connection, and it's associated network settings (ex: SSID, Passcode, Broker, etc) will be in SYSTEM PRORGRAMMING under WiFi Settings.	
COM-40	Control will support LoRa with an optional USB dongle. This will allow for installation with KB LoRa TMS sensors.	
COM-50	Control will support ethernet with an optional USB dongle. This will allow for a hardwire installation and TCP/IP communications.	
COM-60	Control will support local serial communications via RS232 or like with other like Modularm Plus and Pro controls when more than one is connected on the same local installation.	

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COM-70	Control will have an option to echo display statuses from other Modularm Plus & Pro units that are connected together in a local physical installation/location. Ex: Box in a Box control or if a large walk in installation has multiple doors the OEM has an option for have more than 1 display panel on each walk in as well.	
COM-80	There will be an API for front panel features and their associated configuration parameters that can be bi-directionally configured and communicated via the communication means outlined in this document	
COM-90	The unit will support a on-board real time clock (RTC) for event driven features like defrost. The RTC setting will be in system programming. When connected to a local network the clock will sync to the network time.	

4 MECHANICAL REQUIREMENTS

Requirement Number	Description	
ME-00	4.1 MECHANICAL	
ME-05	The PCBA requirements will be a new design with requirements as outlined. The existing functionality of the 75LCT will be used a base for features & functions.	
ME-10	The PCBA will have on board terminals similar to .250 tabs or screw down tyle Phoenix connectors for a secure wire connection.	
ME-20	Enclosure and mechanical footprint to be similar or ideally smaller/shallower then KE2THERM control which is 3.5in wide by 11in high by 19/16in deep. KB engineering to confirm if the 5in LCD can fit 3.5in wide.	
ME-30	Assembly must comply with NEMA splash zone for accidental washdown with cleaning hose or condensing water forming from conduit lines and poor insulated areas.	
ME-40	The USB socket will be accessible from the front panel via a NEMA bulkhead style socket and tethered cap.	
ME-50		

5 PERFORMANCE REQUIREMENTS

Requirement Number	Description	
REG-00	5.1 REGULATORY APPROVALS	
REG-10	Unit will support KB standards for EU60730 and like	
REG-20	Unit will support any addendums for refrigeration controls	
REG-30		
ENV-00	5.2 ENVIRONMENTAL	
ENV-10	The control will be rated for a maximum ambient temperature of 158°F / 70°C, 95% maximum. non-condensing humidity	
ENV-20	The control will operate in a stable condition for standard ESD discharge of up to 25KV from the front panel (when installed in the appliance)	
ENV-30	The control will operate in a stable condition for standard EMF, Emissions, (allowed by agency), in its surroundings when installed in the appliance	

6 OVERLAY AND LABEL REQUIREMENTS

Requirement Number	Description	
OL-00	6.1 OVERLAY	
OL-10	The control will have a mylar overlay with any necessary branding graphics	
LAB-00	6.2 LABELS	
LAB-10	Standard KB product labeling with serial number, date code, ECL revision, agency listings.	
LAB-20	Master pack boxes to the customer will also be labeled on the outside of the box for receiving/inspection	

7 SOFTWARE REQUIREMENTS – MODULARM PLUS

Note: Various logic at the time this document is written will require additional research with respect to refrigeration control, and details will be added as the logic becomes known. Specifically Sections 7.3 & 8.2 as an example.

Requirement Number	Description	
DISP-00	7.1 DISPLAY	
DISP-10	Display will be configured for portrait operation	
DISP-20	There will be a new icon added to the home screen called INFO. This will be freehand text field installers or OEMs can add pertinent information for a service technician. Such as: OEM contact information, refrigeration type, date of last service, date of install, etc.	
DISP-30	There will be an option in system programming to display and transit to KorrekTemp 1/10ths of a degree – both in degrees F & C for the walk in air temperature. Enduser can select if desired. (Ex: 24.7 Degrees C). This is required for certain applications used in product storage, hospital, or lab applications.	
AUDIO-00	7.2 AUDIO	
AUDIO-10	Audio alerts and key beeps will follow current 75LCT patterns	
AUDIO-20	Audio will have typical loudness of ~90db at ~3 feet from the control when mounted in the appliance panel.	
COMP-00	7.3 COMPRESSOR CONTROL	
COMP-10	There is a logic dependency with the compressor output and the Evaporator Fan output that should be noted for logic development.	
COMP-20	The unit will support common compressor control with minimum on times, minimum off times, in programming mode. Consider Multi Evaporators to 1 condenser as an example from VOC from an OEM.	
COMP-30	The unit will support common adaptive logic for controlling the refrigeration air temperature using all three temperature sensors (walk in air temp, condenser coil input side, condenser coil output side.	
COMP-40	When the door is open, the minimum on time must be satisfied before the compressor will turn off.	

COMP-50	When the door is closed, the minimum off time must be satisfied before the compressor is turned back on.	
COMP-60	When Lead Lag is set in the front panel there will be an option for Rapid Cool in programming mode. When Rapid Cool is set/enabled, an icon will appear on the home screen in idle mode called Rapid Cool. When pressed this will enable both SYSTEM A and SYSTEM B refrigeration systems simultaneously to rapidly cool down the temp of the walk in. This will have a parameter in programming mode to either be a temp band from setpoint or a time interval for Rapid Cool to be enabled.	
COMP-70	There will be an option for Rapid Cool to be enabled after the defrost cycle is met.	
COMP-80	Any dependencies of compressor run times, etc that are refrigerant dependent will require a system program parameter of "Refrigerant Type" to be added and any adjustments to temperature control logic will be managed via a table with common commercial refrigerant types used in walk in refrigerators globally. Example: R-454B refrigerant fulfills the EPA mandate for manufacturers to use refrigerants that are better for the environment, with only 25% of the global warming potential (GWP) of the previous refrigerant R-410A. Moving to low-GWP refrigerants is required for all commercial HVAC equipment by January 1, 2025.	
EVAP-00	7.4 EVAPORATOR FAN OUTPUT	
EVAP-10	The evaporator fan will have a programmable parameter in programming mode to interrupt the Fan when the door is open. When this is set in programming as enabled, COMP-30 and COMP-40 will apply.	
EVAP-20	There will be a programmable delay parameter in programming that supports COMP-50, when this time interval is met the evaporator FAN will turn back on and the compressor will turn back on in an attempt to keep the room cool and temps met.	
EVAP-30	There will be a setting in system programming to select either a fixed speed fan output or a variable speed fan output.	
DEFROST-00	7.5 DEFROST CONTROL	
DEFROST-10	The defrost control will be event driven based on the RTC. The user can configure the events based on time of day. There will be a maximum of 4 events in a 24 hour period	
DEFROST-20	Defrost control will support common modes of defrost logic found in refrigeration controls that are both time based events for keeping the compressor off, or temp based threshold values for keeping the compressor off. Logic shall support: Air, Electric, or Hot Gas modes of defrost.	
DEFROST-30	The evaporator fan shall follow common logic modes of operation in defrost mode (Fan Mode – Defrost On/OFF, Pulse or cycle based on temp as an example)	
DEFROST-40	Defrost cycle will complete automatically and resume temperature control	
DEFROST-50	There will be an option in programming mode to show "Defrost in Process" in the display when in defrost mode.	
DEFROST-60	There will be a Demand Defrost mode with programmable blackout times.	
DRHEAT-00	7.6 DOOR HEATER CONTROL	
DRHEAT-10	There will be an option in programming mode to enable the door heater output.	
DRHEAT-20	This can be time based, humidity based, temperature based, or continuous on via a variable voltage to a remote SSR device	
HUMID-00	7.7 HUMIDITY MEASURING	
HUMID-10	The control will have an option in programming mode to sense humidity.	
HUMID-20	When enabled the home display will show the humidity value.	
HUMID-30	When enable in programming mode, there will also be a programmable threshold value to alarm if the humidity levels are too high.	

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HUMID-40	Use of common refrigeration logic should be utilized to measure the humidity and factors that into the refrigeration compressor cycle, coil temps, etc.	

8 SOFTWARE REQUIREMENTS – MODULARM PRO

Requirement Number	Description	
CO2-00	8.1 CO2 SENSING	
CO2-10	The control will monitor the CO2 levels inside the walk in for spoilage detection, employee safety, etc.	
CO2-20	There will be an option in system programming along with a threshold value setting/range.	
CO2-30	When levels are too high, the display will show a CO2 High alarm and trigger an error to the KorrekTemp dashboard for alerting.	
EEV-00	8.2 ELECTRIC EXPANSION VALVE (EEV) CONTROL	
EEV-10	There will be a complimentary EEV to work in conjunction with the adaptive compressor control logic to gain energy efficiencies of the refrigeration system(s). This should be ideally an off the shelf valve if possible to avoid barriers to market.	
EEV-20	There will be compressor control logic to control this valve with respect to compressor control, coil temps, air temp of walk in, etc. that needs to be created or otherwise considered common for refrigeration control which is similar to logic known as “Superheat” in refrigeration applications. This will support Unipolar and Bi-Polar valves.	
EEV-30	Logic needs to be defined for Lead-Lag applications and the use of 1 or more EEV’s in those applications.	
HUMIDFR-00	8.3 HUMIDIFIER CONTROL	
HUMIDFR-10	When the unit is sensing humidity there will be an option to control a humidifier output to work in conjunction to measuring humidity in section 7.7. When Humidity control is set to YES in system programming there will be a threshold value to maintain with the output.	
HUMIDFR-20	It will enable a humidifier output to assist in creating humidity. When the humidity level is met it will turn off the output.	
HUMIDFR-30	There may need to be a hysteresis value associated with this output.	
REFLEAK-00	8.4 REFRIDGERATION LEAK DETECTION	
REFLEAK-10	There will be an option in system programming to sense different types of refrigerant leaks.	
REFLEAK-20	3 rd party devices will be leveraged and the Modularm control input needs to support common hardware interface(s). Use of multiple style sensors based on refrigerant type will be satisfactory if this is the common way of the industry.	
VENT-00	8.5 VENT OUTPUT	
VENT-10	There will be a system programming option for a motorized vent option	
VENT-20	When enabled it will open during a high CO2 level automatically	
VENT-30	When enabled it will open during a refrigeration leak error/alert	
VENT-40	When enabled, it can open during the defrost cycle to assist in defrost mode.	

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VENT-50	When enabled, it can open if humidity levels are too high for an extended period of time interval set in system programming.	
AUTODET-00	8.6 AUTODETECTION	
AUTODET-10	Control will auto detect and self-configure in system programming IP-1, IP-2, MD-1 peripherals when connected on boot up. These can be reset in system programming to factory restore if a user accidentally detects these or should they choose to relocate the control to another walk in.	
AUTOSET-00	8.7 AUTOSET PROGRAMMING	
AUTOSET-10	Control will do a first-time boot for the customer/OEM with 2 icons on the home screen: New Install – Press Freezer or Cooler Setup Once the user or installer chooses the appropriate icon/mode the unit will set the configuration to those defaults as a convenience to programming the unit by hand. Refer to the Autoset setting in the 75LC manual for typical default values for hi temp and low temp alarms. Setpoint default will be 0F for freezer, and 38F for coolers unless otherwise specified.	
AUTOSET-20		
FUTURE-00	8.8 FUTURE CONSIDERATIONS	
FUTURE-10	Future AUX output (10amp) for possible UV Sanitation or like applications for foodservice, or isolation valve application as we know more about controlling refrigeration etc.	
FUTURE-20	Future Serial Communications RS485 port for “add-on” boards/functions. Examples may be a current monitoring board to monitor and measure actual demand, and alert locally and via KorrectTemp when performance issues are recorded	
FUTURE-30	There will be a future multipoint Bluetooth transceiver that will collect temps for food probes and communicate back to the Modularm controller. The control will pair to 1 or more temp sensors inside the walk in that may be inserted into food products (ex: Bluetooth sensor monitoring product that was cooked and needs to be chilled like a alfredo sauce, chili, soup, etc). The Modularm unit will serve as a hub for the temperatures to be collected and forward to the KorrectTemp dashboard for monitoring, alerting, auditing/record keeping for HACCP requirements.	
FUTURE-40	There will be a future gateway device for customers to purchase and connect the Modularm Plus or Pro to allow connectivity to common BAS or Modbus communications for large install customers to manage locally when required. This may include connectivity to a grid monitoring application for energy usage such as Encycle as an example. This would plug on the Future-20 RS485 port as an example.	
FUTURE-50		

9 APPENDIX

KitchenBrains Modularm Plus – Adaptive Controller



- Compressor Control Output (30-amp relay)
- Run Time On/Off, Minimum Run Time
- Coil Temperature Sensor (2)
- Evaporator Fan Output (30-amp relay)
- Fan Modes, Min/Max Run Time, Delay Time
- Defrost Output (30 -amp relay)
- Up to 4 defrost cycles/24hrs)
- Defrost Duration Time, Temperature, Lockout
- Defrost Mode (Air/Electric)
- Demand Defrost
- Door Heater Output (10-amp Relay)
- Humidity Sensing
- Rapid Cool Logic (when connected to Lead-Lag Systems)

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- Run Time On/Off, Minimum Run Time
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- Up to 4 defrost cycles/24hrs)
- Defrost Duration Time, Temperature, Lockout
- Defrost Mode (Air/Electric)
- Demand Defrost
- Door Heater Output (10-amp Relay)
- Humidity Sensing
- Rapid Cool Logic (when connected to Lead-Lag Systems)



- CO2 Sensing
- Electric Expansion Valve Control (EEV) (0-12vdc, pulse count)
- Programmable Valve Percentage/Motor Type
- Programmable Superheat Parameters
- Programmable Pressure Cutout/Differentials
- Refrigeration Type
- Humidifier Output (10-amp relay)
- Refrigerant Leak Detection
- Vent Output (10 -amp relay)

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